

Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

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Motivation

Managing environmental liability in risky industries.

The workhorse tool is **strict liability** — in the UST setting via CERCLA, RCRA Subtitle I, and parallel state law: the firm pays for the harm it causes, regardless of fault.

The judgment-proof problem (Shavell 1986, 2004). When firms' *effective* exposure to liability is less than the harm they might cause — through limited assets, corporate-form shielding, or imperfect detection — strict liability under-deters: too much entry, too little care.

Two financial-responsibility (FR) remedies:

- ▶ (A) Asset / bonding requirements
- ▶ (B) Compulsory liability insurance

Shavell's distinguishing question. Can the insurer **contract on care- and risk-relevant attributes**?

- ▶ Yes $\Rightarrow$ (B) dominates (A): the premium itself varies with care, doing the incentive work.
- ▶ No $\Rightarrow$ (B) can be worse than (A): moral hazard dilutes care.

The Setting: USTs and the Texas 1998 Reform

Research Questions.

- ▶ **Q1.** Does moving from flat-fee public insurance to risk-rated private insurance change firm behavior on closure, replacement, and confirmed-release outcomes?
- ▶ **Q2.** How much of the first-best welfare gain does the risk-rated regime capture, and what determines the residual gap?

Federal Financial Responsibility mandate (1988) — every UST operator must carry \$1M coverage. The *form* of that coverage varies by state:

- ▶ **38 states** — public trust fund; uniform per-tank fee. *The insurance contract does not condition on risk attributes.*
- ▶ **Texas, Dec 1998** — fund closed; operators forced into the **private risk-rated market**. *Premiums condition on age, wall, piping, leak-detection.*

Tank attributes are recorded in EPA/state inventories under both regimes; the reform changes whether the insurance contract prices on them.

Contribution

Shavell instrument choice & financial responsibility (theory). Shavell (1982, 1986, 2004); Yin, Pfaff & Kunreuther (2011)

Empirical financial-assurance — asset / bonding leg. Boomhower (2019); Ho, Hsu, Cha & Rivera (2018); Yin, Kunreuther & White (2011)

Insurance pricing in environmental settings. Mulder (2022); Yin, Pfaff & Kunreuther (2007, 2011)

Dynamic structural welfare under environmental regulation. Fowlie, Reguant & Ryan (2016); Muehlenbachs (2015); Blundell, Gowrisankaran & Langer (2020)

This paper. First firm-level empirical test of the **pure pricing channel** of compulsory liability insurance — the **(B)** leg of the Shavell FR menu. Distinguished from (A)-leg work (Boomhower, Ho et al.) and from priced-insurance settings with confounding information channels (Mulder). Pairs reduced-form evidence on closure, replacement, and confirmed releases with a dynamic structural welfare decomposition that bounds the gap to first-best.

Empirical Strategy

- ▶ **Quasi-experiment.** Dec 1998: TX closes its uniform-premium fund; operators forced into the risk-rated private market.
- ▶ **Sample.** TX + 17 control states (a subset of the 38 uniform-premium states with usable admin data); $\sim 9.8\text{M}$ tank-years.
- ▶ **Reduced form.** DiD with facility FE and cell $\times$ year FE.
- ▶ **Structural.** Dynamic discrete choice (maintain / exit / replace) estimated by NPL; recovers $(\kappa, K, \gamma_p, \gamma_r)$.
- ▶ **Counterfactuals.** Subsidy, Pigouvian, mandate. Welfare = firm surplus – damage PV – outlay PV.

Main Findings

Reduced form (*alive-at-reform sample, $N \approx 9.8M$ tank-years*)

- ▶ TX $\times$ Post raises annual tank-closure probability by +1.58 **pp** on a 2.0% baseline
- ▶ 4 $\times$ larger effect in pre-1989 vintage (+3.1 pp) — the high-risk SW-heavy stock responds most
- ▶ LUST discovery falls -1.27 **pp** on a 1.73% baseline; both background and inspection-triggered drop
- ▶ Conditional on closing: +53% relative shift to the **replacement** margin (12.0% $\rightarrow$ 18.3%); aggregate exit volume unchanged

Structural (4-parameter DDC) (*in units of annual revenue R , with $R \approx \$10k$ as a calibration anchor*)

- ▶ $\hat{\kappa} \approx 26 R$, $\hat{K} \approx 5 R$, $\hat{\gamma}_p = -0.29$, $\hat{\gamma}_r = 0.16$
- ▶ All three counterfactuals (Subsidy / Pigouvian / Mandate at age 25+) reduce social welfare at $E = \$17k$ (Marcus, 2021); optimal K subsidy $s^* \approx 10\%$, far below 50%

A Dynamic Model of Tank Closure

The Agent's Decision

The Choice. Each period t , the facility chooses $d_t \in \{\text{Maintain, Close}\}$.

The Trade-off.

- ▶ **Maintain:** Earn net revenue, pay insurance premiums.
- ▶ **Close:** Exit and recover scrap value κ (liquidity of land).

Flow Utility (Maintain).

$$u(a) = \underbrace{R}_{\text{Net Revenue}} - \underbrace{C_j(a)}_{\text{Insurance Premium}}$$

Simplifying assumption (full coverage): environmental risk enters as an *ex-ante* premium cost, not an *ex-post* shock.

The Dynamic Problem

Bellman equation.

$$V(a) = \max \left\{ \underbrace{u(a) + \beta \mathbb{E}[V(a') \mid a]}_{\text{value of maintaining}}, \underbrace{\kappa}_{\text{scrap value}} \right\}$$

Continuation value. The value of keeping the tank active today to preserve the option of operating tomorrow:

$$V_{\text{cont}}(a) = u(a) + \beta \mathbb{E}[V(a') \mid a]$$

Optimal stopping rule. Close if $V_{\text{cont}}(a) < \kappa$.

The Policy Lever: Insurance Regimes

1. Uniform Premium Pooling (F) — 38 states.

- ▶ Premium: constant $\bar{P}$, regardless of age or risk.
- ▶ Marginal cost of aging: $\partial C / \partial a = 0$. No incentive to close aging tanks.

2. Risk-Based Pricing (RB) — Texas post-1999.

- ▶ Premium: rises with the leak hazard, $C(a) = \lambda \cdot h(a) \cdot L$.
- ▶ Marginal cost of aging: $\partial C / \partial a > 0$. Aging tanks become increasingly costly.

What's the First Best?

Social planner's objective. Internalize remediation costs (L) and health externalities (E):

$$u_{\text{SOC}}(a) = R - h(a) \cdot \left(\underbrace{L}_{\text{Remediation}} + \underbrace{E}_{\text{Health}} \right)$$

Why risk-based pricing falls short.

- ▶ Actuarial premiums reflect insurer claims data → only remediation costs (L)
- ▶ Health damages (E) generate no third-party claims: contamination is unobservable (Marcus, 2021)
- ▶ $\Rightarrow$ Even fair risk-based pricing leaves E externalized

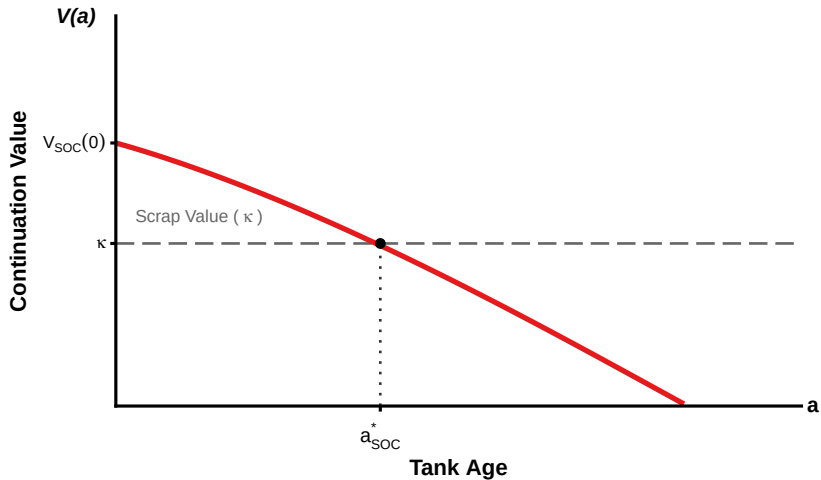
Implication. $V_{\text{RB}}(a) > V_{\text{SOC}}(a)$ — RBP induces *some* closures, but not enough.

Three Facts to Keep in Mind

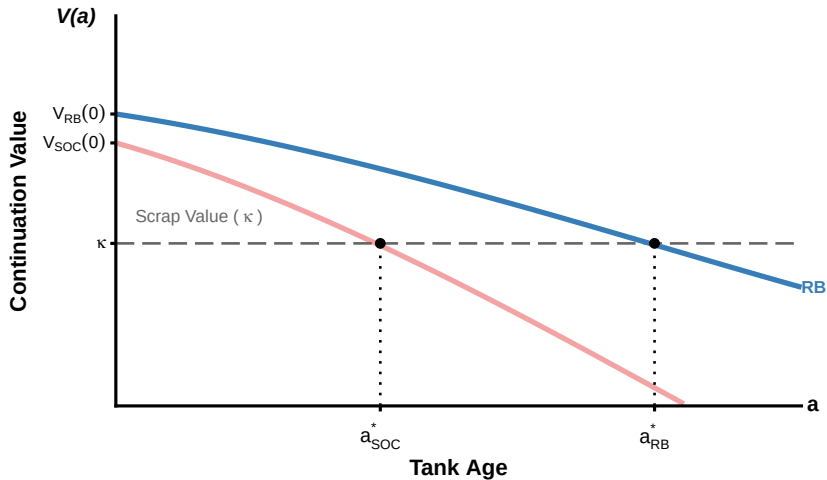
- 1. Asset depreciation.** Rising leak risk drives tank value strictly down in age ($V'(a) < 0$). The regime determines whether private incentives align with this social decay.
- 2. Inefficient retention (Uniform Premium).** Static costs $\bar{P}$ imply zero marginal cost of aging. Firms ignore risk and retain assets longest: a_{UP}^* .
- 3. Partial correction (Risk-Based).** Rising premiums $h(a) L$ imply positive marginal cost of aging. Closure accelerates to a_{RB}^* , but still $> a_{SOC}^*$ due to unpriced health damages.

$$a_{SOC}^* < a_{RB}^* < a_{UP}^*$$

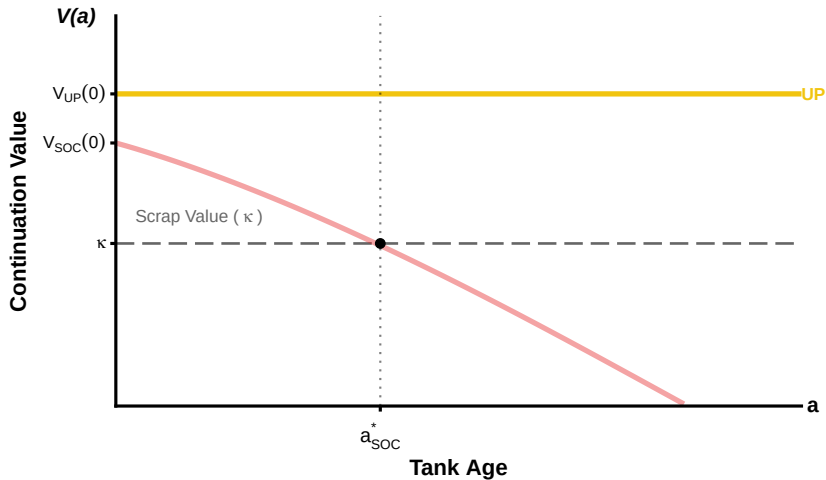
Social Optimum Exit Threshold



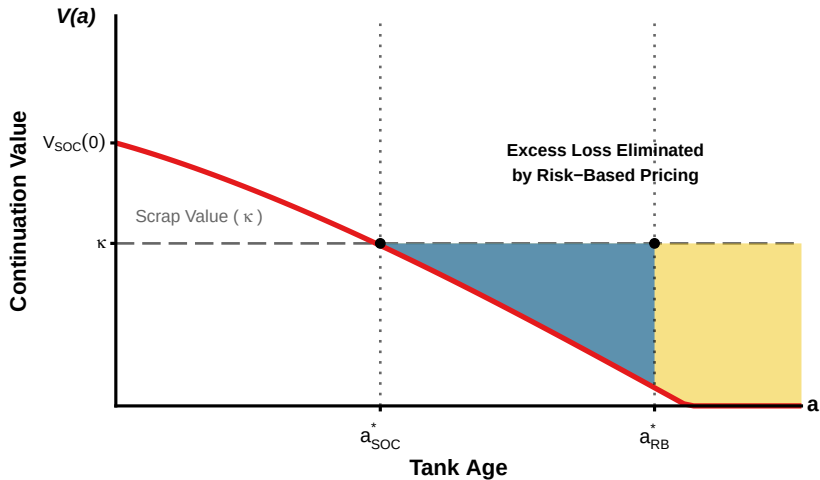
Risk-Based vs. Social Optimum



Uniform Premium vs. Social Optimum

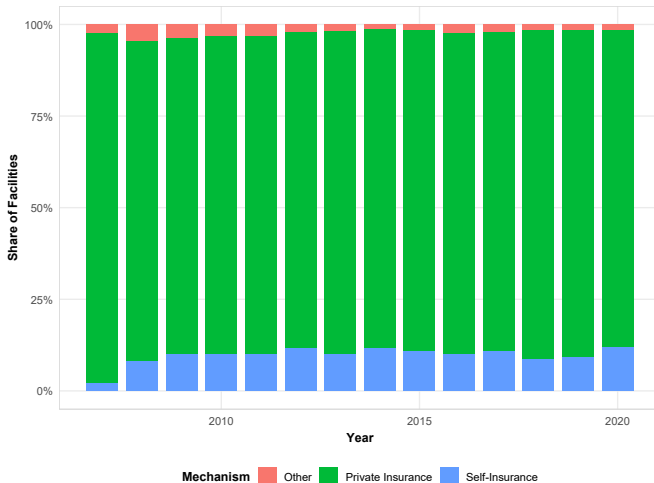


Deadweight Loss Across Regimes

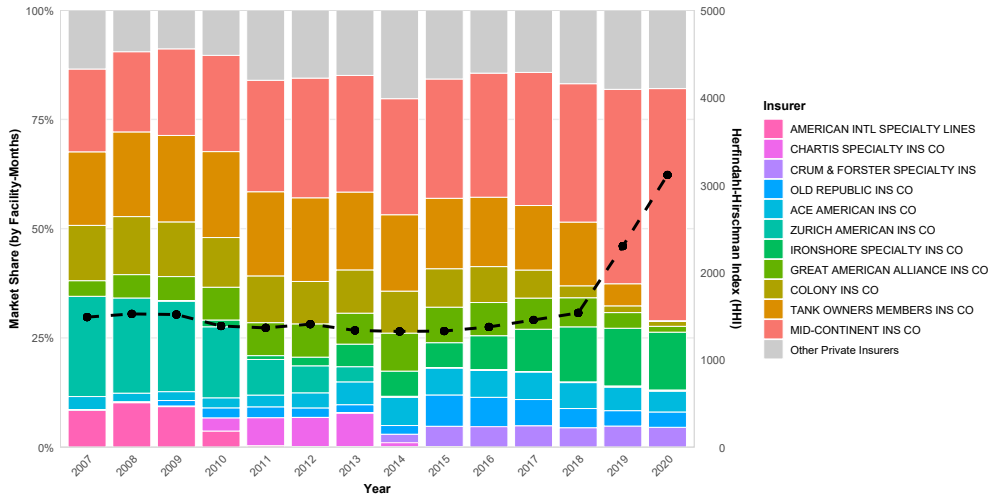


Data & Descriptive Evidence

FR Compliance: TX is a Private-Insurance Market



Market Structure: Concentrated and Stable



Premium Construction

Texas — rebuilt from Mid-Continent Casualty rate filings (SERFF, 2006–2024):

$$P_{ijt} = \text{Base}_t \times \text{ILF}_{it} \times \left(1 + \sum_k \lambda_{k,t} X_{ijt,k}\right)$$

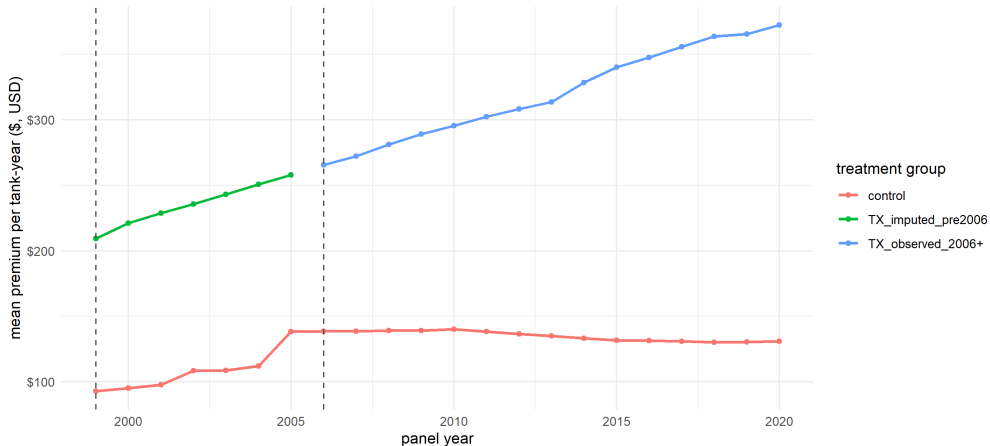
- ▶ Base_t : per-tank base rate, four filing periods
- ▶ ILF_{it} : increased-limit factor; fix coverage at federal-minimum \$1M/\$1M
- ▶ X_{ijt} : tank age, wall, piping, leak-detection technology
- ▶ $\lambda_{k,t}$: additive credits/debits from the filed schedule

Control states — state financial-responsibility fund records: a per-tank annual assessment, largely flat across years and tank characteristics.

Premium Levels: Texas vs. Control States

Mean per-tank premium by year x treatment group — RAW DATA

TX_imputed_pre2006 = engine-imputed Mid-Continent premium for 1999-2005



Risk Prediction: Setup

Estimand. Annual first-release hazard in the never-yet-leaked risk set:

$$h(t \mid x) = P(\text{first confirmed release in year } t \mid \text{no prior release, } x)$$

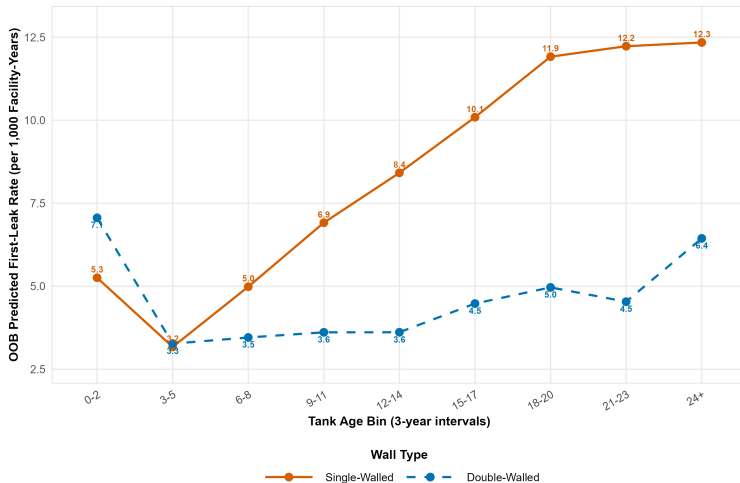
In plain words: given what an underwriter can observe at the start of a policy year, what's the probability of a claim?

Model. Penalized logistic regression on the pre-treatment panel:

- ▶ x : 3-year age bins fully interacted with wall construction; state FE; fuel and portfolio controls
- ▶ Lasso variable selection; inverse-frequency weights for the 0.70% base rate
- ▶ Out-of-sample CV (10 folds); post-hoc recalibration to the empirical event rate

Sample. 3.93M pre-treatment facility-years; 27,455 confirmed first-release events. $\hat{h}(x)$ feeds the next figure and the structural-model hazard $h(a_t, W_i)$.

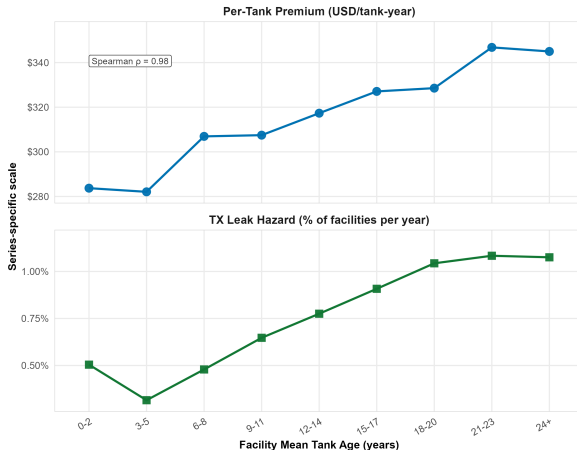
Single-Walled Tanks Drive the Age-Hazard Gradient



Premiums Track the Risk We Can Predict

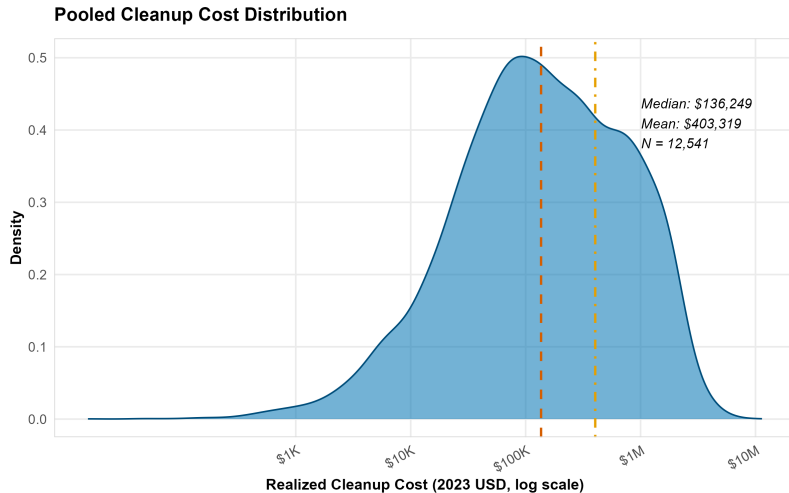
Premium Schedule Tracks Texas-Estimated Leak Hazard

Per-tank premium (TX Mid-Continent, 2006-2014) and TX leak hazard (01n elastic net OOB) both rise across age bin



Notes: Per-tank premium = mean(tank_premium) across TX Mid-Continent tank-months in each bin. TX leak hazard = mean of row-level

Cleanup Cost Distribution



Reduced Form Evidence

Sample Descriptive Statistics

	Texas	17 control states
<i>Panel size</i>		
Facility-years (incumbent, 1985–2018)	870,341	3,782,424
Tank-years (active-at-treatment)		9,761,982
<i>Tank stock at Dec 22, 1998 (per facility)</i>		
Facilities	25,985	114,304
Mean tanks	2.78	2.74
Single-walled share (tank-weighted)	89.1%	89.5%
Mean oldest tank age, years	15.32	18.09
Mean total capacity, gal	22,585	62,405
Gasoline facility share	66.1%	75.4%
<i>Vintage composition (facility counts, full panel)</i>		
Pre-1989 vintage (faced EPA phase-in)		96,923
Post-1988 vintage		42,523

Estimating Sample and Specification

Static DiD.

$$Y_{it} = \beta (\text{Texas}_{f(i)} \times \text{Post}_t) + \alpha_i + \delta_{ct} + \varepsilon_{it}$$

Event study.

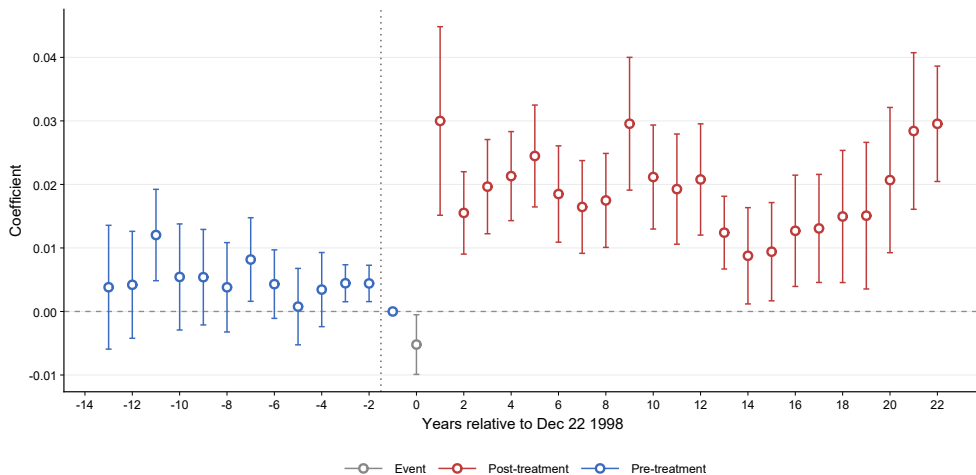
$$Y_{it} = \sum_{\tau \neq -1} \beta_{\tau} \cdot (\text{Texas}_{f(i)} \times \mathbf{1}[t - 1998 = \tau]) + \alpha_i + \delta_{ct} + \varepsilon_{it}$$

- ▶ α_i : facility FE. δ_{ct} : cell $\times$ year FE with $c = \{\text{wall} \times \text{fuel} \times \text{capacity} \times \text{install cohort}\}$
- ▶ *Identifying variation*: within a single tank-vintage cell, in the same calendar year, β is the change in the Texas-vs-control closure-rate gap before vs. after Dec 22, 1998

ATT on Tank Closure: +1.58 pp on a 2.0% Baseline

	(1) Raw (1)	(2) + Fac FE (2)	(3) + Controls (3)	(4) + Cell FE (4)
Constant	0.0203*** (0.0022)			
did_term	0.0106*** (0.0022)	0.0338*** (1×10^{-6})	0.0350*** (0.0015)	0.0158*** (0.0040)
mandate_release_det			-0.0039 (0.0035)	
mandate_spill_overfill			0.0048* (0.0025)	
mandate_integrity			0.0085*** (0.0026)	
Observations	0.761.082	0.761.082	0.761.082	0.761.044

Dynamic Closure Response in Texas



Heterogeneity: Pre-89 Vintage Drives the Effect

	(1) Raw (1)	(2) + Fac FE (2)	(3) + Controls (3)	(4) + Cell FE (4)
Constant	0.0103*** (0.0011)			
did_term	0.0048*** (0.0011)	0.0117*** (0.0028)	0.0114*** (0.0029)	-0.0067** (0.0031)
pre89_cohort	0.0135*** (0.0023)	0.0671*** (0.0062)	0.0696*** (0.0060)	
did_term × pre89_cohort	0.0094*** (0.0023)	0.0382*** (0.0029)	0.0360*** (0.0029)	0.0309*** (0.0065)
mandate_release_det			-0.0114** (0.0040)	
mandate_spill_overfill			0.0009 (0.0028)	
mandate_integrity			0.0044 (0.0030)	
Observations	9,761,982	9,761,982	9,761,982	9,761,044
R ²	0.00249	0.04685	0.04750	0.07772
Within R ²		0.01392	0.01460	0.00065
panel_id fixed effects		/	/	/

ATT by install year →

LUST Discovery: Both Channels Fall

Outcome	Pre-reform ctrl mean	TX $\times$ Post
Any leak found this year	1.73%	−1.27 pp ^{***} (0.0032)
<i>By how the leak was discovered:</i>		
Routine monitoring (no closure in $\pm 60d$)	1.16%	−0.76 pp ^{***} (0.0021)
Closure-inspection finding (closure within 60d)	0.61%	−0.51 pp ^{**} (0.0017)
N facility-years	4,650,240	
Sample	Incumbent (alive at Dec 22, 1998)	
FE	Facility + cell $\times$ year; cluster by state	

What Closing Firms Do: Composition Shifts to Replacement

Outcome (conditional on closing)	Pre-reform ctrl mean	TX $\times$ Post
Facility closes all tanks this year	70.9%	+6.5 pp ^{***} (0.019)
Has a permanent closure (no replace)	88.3%	-6.2 pp ^{***} (0.012)
Has a replacement closure (upgrade)	12.0%	+6.3 pp ^{***} (0.013)
Any DW tank installed	4.1%	-1.4 pp [*] (0.007)
Net tank change (count)	-1.905	+0.020 (0.085)
Capacity change (gal)	-100,394	-1,397 (5,715)
<i>N</i> closing facility-years	59,107	
Sample	Incumbent, any_closure = 1	
FE	Facility + make_model_fac $\times$ year; cluster by state	

Structural Model

Model Setup

State. $s_t = (a_t, W_i, R_i, \varepsilon_t)$ — 9 age bins $\times$ 2 wall types $\times$ 2 regimes = 36 states. Actions: maintain / exit / replace.

Flow utility (annual net revenue R normalized to 1; *not directly observed* — the model identifies θ in units of R).

$$u^m(s) = 1 - \gamma_p p_t - \gamma_r h(a_t) L$$

$$u^e(s) = \kappa \quad (\text{scrap value})$$

$$u^r(s) = -K \quad (\text{install new DW tank})$$

Calibrated upstream (*outside the model*): premium p_t , hazard $h(a_t)$, liability L — all from the descriptive-evidence section.

Estimated jointly (*inside the model*): $\theta = (\kappa, K, \gamma_p, \gamma_r)$. NPL match of choice shares across (a, W, R) cells.

Identification

Each parameter is identified by a distinct moment in the choice data:

- ▶ κ : the unconditional exit rate $P(\text{exit} \mid s)$
- ▶ K : the replace-vs-exit share among closing tanks
- ▶ γ_p : cross-cell response of choice shares to premium variation in (a, W, R)
- ▶ γ_r : divergence in age-closure hazard between uniform-premium and risk-based regimes

NPL (Aguirregabiria & Mira 2007); AM-2002 profile-likelihood SEs.

Structural Estimating Sample

Unit of observation. Facility-year. One row per facility per year, 1999–2018.

Sample. Texas facilities, post-reform period. The structural model fits the *equilibrium* of the risk-based regime — TX is the regime that's actually priced.

- ▶ Roughly 26k TX incumbent facilities over 20 years $\approx$ 520k facility-years
- ▶ After dropping rows with missing premium or hazard inputs: $\approx$ 430k facility-years
- ▶ Choices observed: maintain / exit / replace; mapped onto 32 state cells (a, W, R)
- ▶ Premiums p_t from Mid-Continent rate-filing reconstruction; hazards $h(a_t)$ from the first-release-prediction model

Parameter Estimates

	Observed TX 2006+, controls 1999+	Extended (2000+) TX 2000+, controls 2000+
κ_{exit} (\$)	\$261,222 (\$306)	\$257,702 (\$310)
K (\$)	\$51,779 (\$142)	\$52,658 (\$146)
γ_{price}	-0.293 (0.049)	-1.142 (0.048)
γ_{risk}	0.164 (0.003)	0.153 (0.003)
$\log L$	-268,481	-247,614
N obs	2,282,735	2,300,780

SEs are AM (2002) profile-likelihood; bootstrap SEs deferred for a future pass.

Estimates in Economic Terms

Reminder: all parameters are in units of annual net revenue R . Dollar translations below use a calibration anchor of $R \approx \$10\text{k/yr}$ per facility

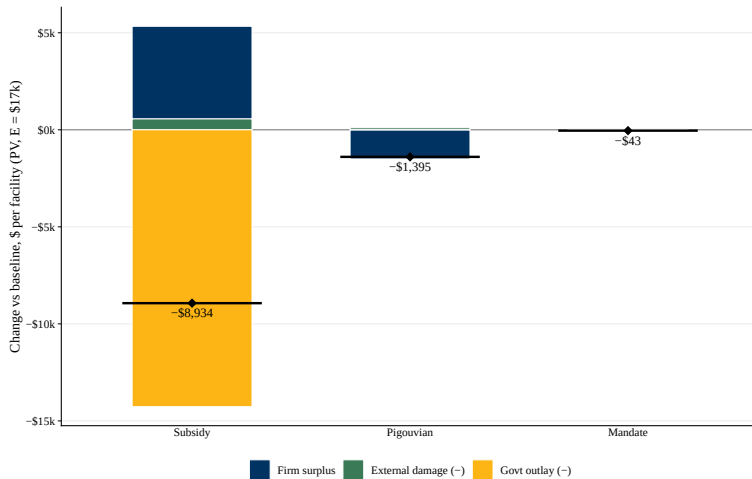
- ▶ $\hat{\kappa} \approx 26 R$ (about \$261k under the calibration). Net scrap value when a facility exits; large in absolute terms, consistent with land-value liquidation net of remediation liability.
- ▶ $\hat{K} \approx 5 R$ (about \$52k). Replacement cost for a new double-walled tank; in line with industry estimates.
- ▶ $\hat{\gamma}_p = -0.29$. a one-unit ($\approx 1 R$) increase in premium reduces the maintain-flow value by 0.29 units.
- ▶ $\hat{\gamma}_r = 0.16$. The point estimate is small relative to a fully-internalized benchmark ($\gamma_r = 1$).

Counterfactual Policy Scenarios

Policy	Mechanism
Subsidy	$K \rightarrow 0.5 K$ — replacement cost halved
Pigouvian	Add $h(a) \cdot E$ to maintain cost; $E = \$17,000$ (Marcus, 2021)
Mandate	Force $P(\text{maintain}) = 0$ for $a \geq 6$ (age 25+)

What we do for each policy. Recompute firms' optimal choices under the new price/cost regime, let the fleet settle into its long-run age-and-wall distribution, then add up the present-value impact on three groups: firms (operating profits), the environment (avoided cleanup and health damages), and the government (subsidies paid out).

Welfare Decomposition



Counterfactual Welfare Summary

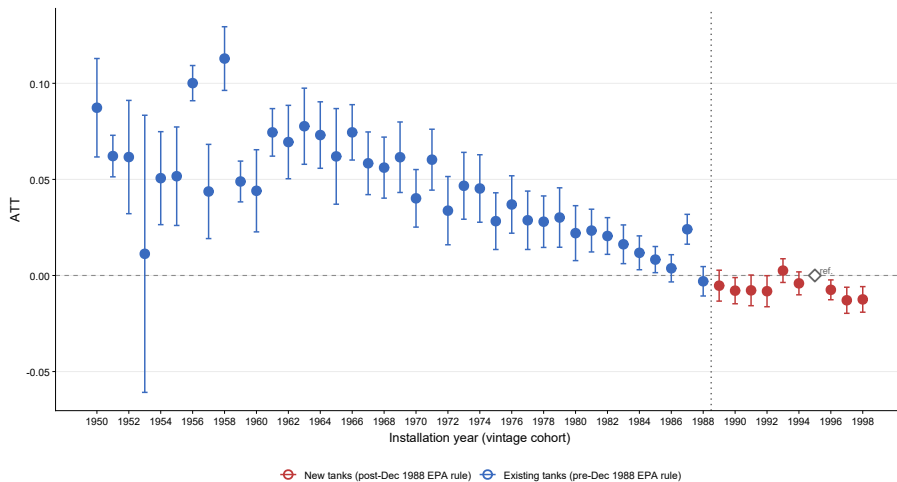
	$\bar{P}(M)$	$\bar{P}(E)$	$\bar{P}(R)$	Firm Surplus	Social Welfare	ΔW
Baseline	98.0%	1.7%	0.2%	\$307,724	\$305,344	$\pm \$0$
Subsidy	96.2%	1.1%	2.8%	\$312,498	\$296,409	-\$8,934
Pigouvian	97.7%	2.0%	0.2%	\$306,225	\$303,948	-\$1,395
Mandate	98.0%	1.8%	0.2%	\$307,633	\$305,301	-\$43

All three CFs reduce social welfare at $E = \$17k$. Subsidy hurts most: govt outlay outweighs firm-surplus gain. Mandate roughly neutral.

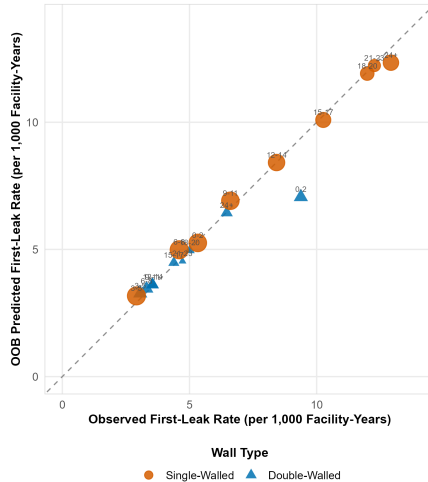
Takeaways

1. **Risk-based pricing does shift behavior.** A +1.58 pp annual closure ATT — about 4× larger in the pre-1989 SW-heavy stock that the schedule actually prices.
2. **The shift is on the composition margin, not exit volume.** Conditional on closing, the **replacement** rate nearly doubles (12% → 18%); net tank counts and capacity barely move. LUST discovery falls through both background and inspection-triggered channels.
3. **Welfare under the priced regime is still below first-best.** All three counterfactual policies — K subsidy, Pigouvian surcharge on $h(a) \cdot E$, age-25 mandate — *reduce* social welfare at $E = \$17k$. RBP is a partial fix; closing the residual wedge requires direct intervention beyond the actuarial price signal.

Appendix: ATT by Installation Vintage



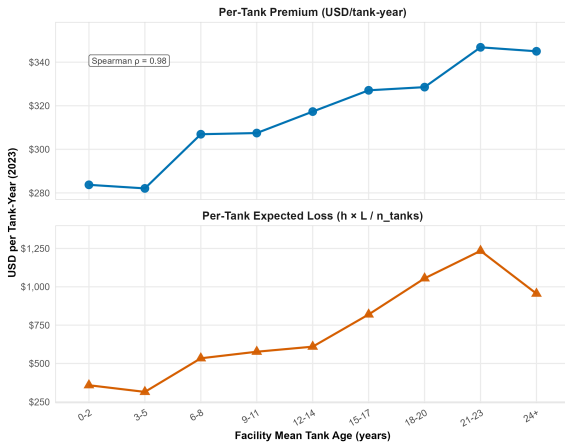
Appendix: Cell-Level Goodness of Fit



Appendix: Premium vs. Per-Tank Expected Loss

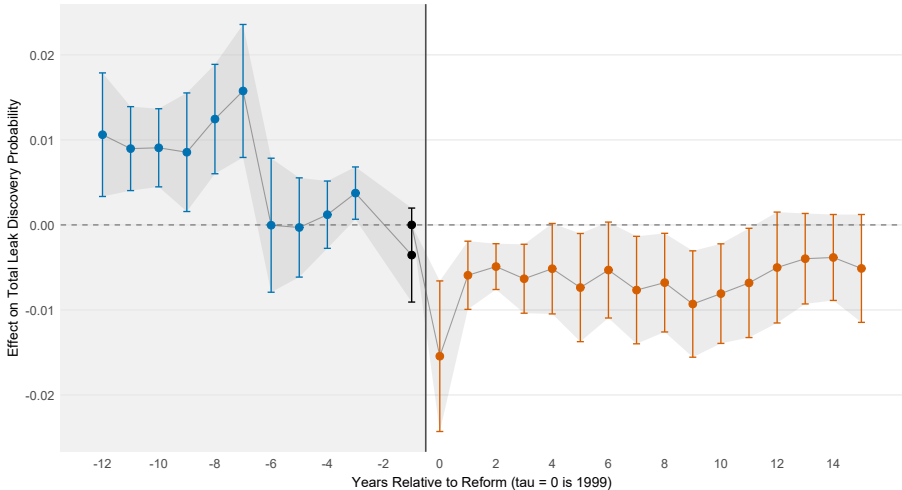
Premium and Constructed Per-Tank Expected Loss Across Age Bins

Backup view: per-tank EL = $\hat{h} \times \hat{L} / n_{\text{tanks}}$, where $\hat{L}$ is a control-state donor curve (no TX cost data exi



Notes: Per-tank premium as in main figure. Per-tank expected loss = $\hat{h} \times \hat{L} / n_{\text{tanks}}$, computed at the tank-month level using ea

Appendix: LUST Event Study — Total Leak Discovery



Appendix: Subsidy Curve

